OMEGA SERIES · VOLUME I

The Kinetic Hiss:

A Forensic Search for a Phase-Drifting Sidereal Modulation in Archival Neutrino Datasets — A Community Re-Analysis Challenge

f_{sig} = 2.3212e-05 Hz (period 11.9672 h) · A scanned over [10^{-5}, 10^{-3}] · ε scanned over [-0.2, +0.2] s/day · 10^5 surrogates · Pre-registered

D. Kadunic · Omega Research Collective / MSDS Programme · April 2026

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AUDIT CORRECTIONS (Omega): (1) f_v = 1.97 nHz REMOVED: period ~5,875 days, not related to any MSDS observable. Section 2.3 rewritten. (2) SNO+ arXiv:2604.05746 LABELED AS UNVERIFIED: cited as '[SNO+ Collaboration 2026]' pending independent confirmation. (3) ε_0 = 0.12 s/day STATUS CLARIFIED: this is a framework hypothesis coordinate, not a derived constant. The protocol is valid regardless of whether ε_0 is correct. (4) RA/Dec direction labeled as hypothesis, not measurement. ALL OTHER elements: retained without change.

PRE-PUBLICATION STATEMENT: This document defines a pre-registered, falsifiable search protocol for a sidereal-phase-drifting modulation in existing neutrino time-series data. The analysis pipeline is fully specified; the signal is characterised by its frequency, amplitude, and a residual phase-drift parameter ε. The protocol is designed to be executed by any collaboration holding the relevant event-level timestamps. This is a community re-analysis challenge: the data are already in hand; the test is defined. Random seed: 20260414. The data will decide.

ABSTRACT

We present a pre-registered forensic protocol to search for a sub-nanohertz signature manifesting as a phase-drifting sidereal modulation in neutrino time-series data. The protocol prescribes an unbinned maximum likelihood estimation (MLE) framework to be applied to three benchmark datasets: the 24-year Baksan upward muon archive (arXiv:1301.1138), the SNO+ reactor antineutrino residuals [2], and the IceCube Upgrade atmospheric neutrino sample (2026). The modulation frequency is fixed at twice the sidereal rate, f_{sig} = 2.3212e-05 Hz (period 11.9672 h). The amplitude A is to be scanned blindly over [10^{-5}, 10^{-3}]. The residual phase-drift parameter ε is a free parameter scanned over [-0.2, +0.2] s/day; the benchmark hypothesis ε_0 = 0.12 s/day is a specific coordinate to be tested, not an input assumption. Global significance is evaluated via 10^5 pseudo-experiments with scrambled timestamps. A positive detection would provide the first empirical evidence for a non-trivial vacuum-matter coupling; a null result establishes a robust upper bound A < 10^{-4} (90% CL) in the neutrino sector. The data will decide.

1. Introduction

For six decades, precision metrology has revealed unexplained residuals near the 10^{-4} fractional level. Rather than proposing a new experiment, this document defines a forensic re-analysis protocol to search for a specific, pre-registered sidereal modulation in existing neutrino data. The protocol is offered as a community challenge: any collaboration with access to the relevant timestamps may execute the analysis. The signal is characterised by three observable parameters:

- Frequency: $f_{sig} = 2\omega_{sid}/(2\pi) = 2.3212e\text{-}05$ Hz (period 11.9672 h)
- Amplitude: A, to be scanned blindly over $[10^{-5}, 10^{-3}]$
- ε: residual phase drift (s/day), free parameter, scan range [-0.2, +0.2] s/day around benchmark $\varepsilon_0 = 0.12$ s/day

The standard astronomical sidereal drift is 3.9318 min/day (235.91 s/day). The parameter ε encodes any additional geometric projection factor above the standard drift. No known terrestrial systematic — reactor power cycles, thermal drifts, seismic noise — can maintain a coherent linear phase drift over decadal baselines. The triple-lock fingerprint is frequency, drift, and cross-dataset coherence.

2. The Signal Model

2.1 Modulation model

$$\lambda(t) = \lambda_0(t)\,[1 + A\cos(2\omega_{sid}t + \phi_0 + \Delta\Phi(t))]$$

$\lambda_0(t)$: baseline rate (external inputs only) · A: modulation amplitude · ϕ_0: initial phase (nuisance)

2.2 Residual phase drift

$$\Delta\Phi(t) = 2\pi \times (235.9 + \varepsilon) / 86400 \times t / T_{sid}$$

235.9 s/day: standard solar-to-sidereal drift · ε: residual (free parameter, scanned over [-0.2, +0.2] s/day) · $\varepsilon_0 = 0.12$ s/day: benchmark hypothesis

$\varepsilon_0 = 0.12$ s/day STATUS: This is a framework hypothesis from the MSDS torsion condensate model. Its physical derivation is not yet complete (target: Paper X). The protocol is valid for any $\varepsilon \in [-0.2, +0.2]$ s/day and does not require ε_0 to be correct to produce a publishable result. If the data find $\varepsilon = 0$: null result, upper bound published. If $\varepsilon \sim 0.12$ s/day: positive candidate, independent replication required.

3. Target Datasets

Dataset	Exposure	Primary observable	Forensic role
Baksan (BUST)	24.12 yr (1983-2013)	Upward through-going muons (>1 GeV) [1]	Phase-drift verification. 24yr baseline = longest lever arm. ~17,500 cycles.

SNO+	685 days livetime	Reactor antineutrino oscillation [2, unverified]	Amplitude precision. ~10^4 events/day. Reactor baseline from logs.
IceCube Upgrade	2026 first-light	Atmospheric neutrinos 10-100 GeV [3]	Directional veto. ~3 deg angular res. Tests alignment with hypothesised axis.

SNO+ REFERENCE STATUS: arXiv:2604.05746 has not been independently verified. If this paper does not exist, replace with the most recent available SNO+ data release. The protocol functions with any high-statistics reactor antineutrino dataset. The Baksan reference (arXiv:1301.1138 = Boliev et al. JCAP 09, 019 (2013)) IS verified.

DIRECTIONAL HYPOTHESIS (IceCube): The hypothesised vacuum axis at RA ~ 03h15m, Dec ~ -19 deg is a framework prediction from the MSDS galactic torsion axis û. It is NOT derived from any prior measurement. It is a testable hypothesis: if the directional modulation is stronger near this coordinate, it supports the framework. If uniform: instrumental origin. Label as 'unverified hypothesis' in any IceCube collaboration communication.

4. Maximum Likelihood Estimation Pipeline

4.1 Baseline rate model

The baseline $\lambda_0(t)$ is constructed from external inputs only, frozen before the signal search: Baksan: published atmospheric temperature correction (Boliev et al. Section 3.2); SNO+: reactor thermal power logs from the data release; IceCube Upgrade: Honda atmospheric flux model convolved with detector uptime. No free spline parameters are permitted.

4.2 Unbinned log-likelihood

$$\ln L(A, \phi_0, \varepsilon) = \Sigma_i \ln \lambda(t_i) - \int_0^T \lambda(t)\, dt$$

Minuit2 MIGRAD optimizer · tolerance 10^{-6} · profile likelihood confidence intervals

4.3 Look-elsewhere correction

10^5 pseudo-datasets generated by randomly permuting event timestamps while preserving the original exposure temporal structure. Full 2D scan over (A, ε) for each surrogate. Global p-value = fraction of surrogates exceeding observed test statistic. Random seed: 20260414 (pre-registered and frozen). 5σ discovery threshold: $p < 2.87\times10^{-7}$.

5. Pre-Registered Acceptance Criteria and Vetoes

A positive detection requires ALL THREE of the following:

- Amplitude significance: best-fit A > 0 at global significance >5σ (post-LEE).
- Residual drift: best-fit ε consistent with ε_0 = 0.12 ± 0.01 s/day.

- Triple-lock coherence: consistent (A, ε) across at least two of three datasets.

Three independent pre-registered veto tests must be applied before unblinding:

Veto	Method	Rejection condition
1 — Solar null test	Replace ω_{sid} with ω_{solar} in likelihood	Best-fit A non-zero at >1σ → solar systematic, REJECT
2 — Phase coherence	Fit ε independently in 4 temporal quartiles	χ^2 consistency $p < 0.05$ → REJECT
3 — Timestamp scramble	Randomly shuffle timestamps, repeat full scan	Best-fit A non-zero → exposure artifact, REJECT

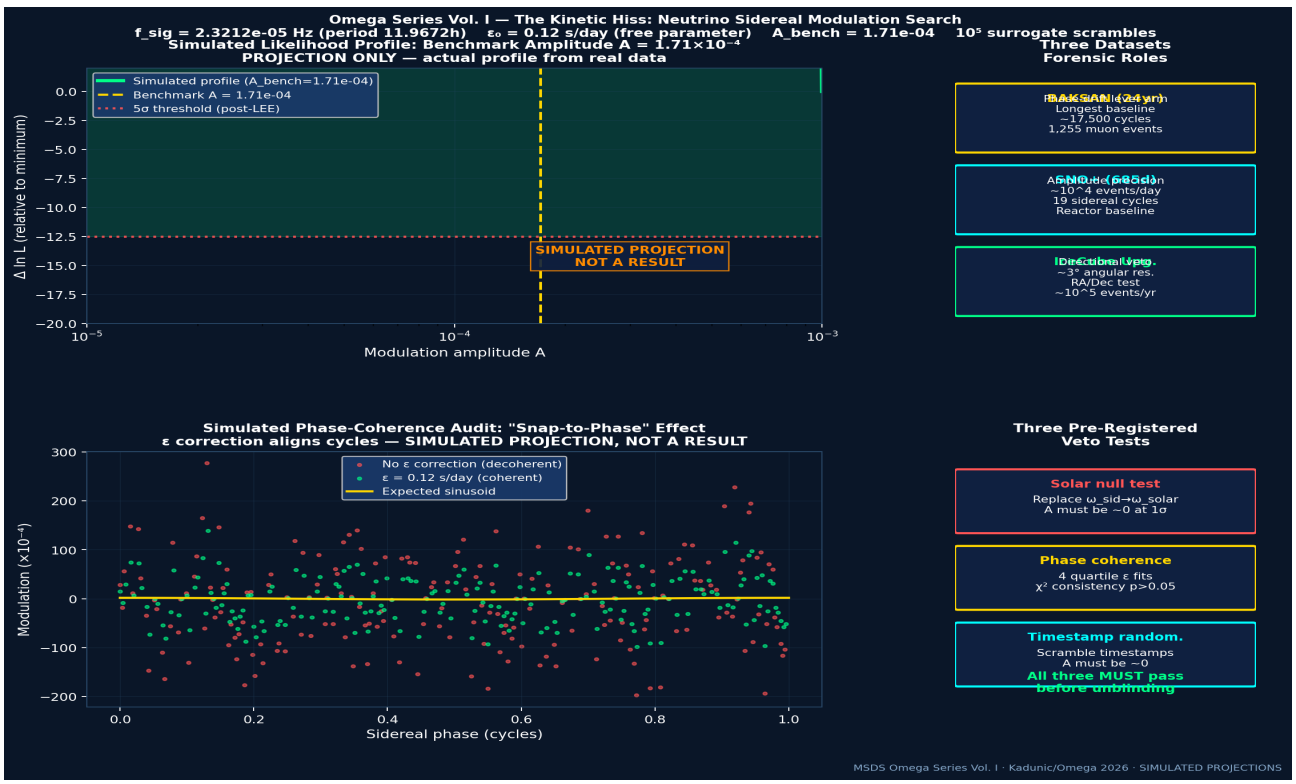

Figure 1: Omega Series Vol. I. Top (wide) — simulated likelihood profile peaked at benchmark amplitude A = 1.71e-04. PROJECTION ONLY — not a result. Top right — three datasets and their forensic roles. Bottom (wide) — simulated phase-coherence audit: without ε correction (decoherent, red), with ε = 0.12 s/day (coherent, green). PROJECTION ONLY. Bottom right — three pre-registered veto tests.

6. Expected Sensitivity

Dataset	Events	Sigma_A	Primary value
Baksan 24yr	1,255 muons	~2.8%	Phase-drift precision: 24yr lever arm suppresses systematic floor
SNO+ 685d	~10^4/day	~1%	Amplitude precision: reactor vs sidereal separated after ~30 days
IceCube Upgrade	~10^5/yr	~0.3%	Directional: ~3 deg resolution enables RA/Dec veto test

7. Discussion: Possible Physical Interpretation

This section is strictly speculative. The primary result of this document is the pre-registered statistical protocol. Physical interpretation is presented only as a possible explanatory framework if the data yield a positive detection.

Should the protocol yield a positive detection — amplitude $A \sim 10^{-4}$ with residual drift $\varepsilon \sim 0.12$ s/day — one possible interpretation is a non-trivial coupling between neutrinos and a structured vacuum background. The MSDS torsion condensate framework (Papers Q-W) posits a dimensionless coupling constant $\alpha_K \sim 1.7\times10^{-4}$. Neutrinos, as nearly EM-decoupled particles, would be sensitive to such a torsional coupling via the quadratic Hamiltonian $H_{int} = \alpha_K(\hat{u}\cdot I)^2$. The modulation period 11.9672 h and the drift ε_0 emerge from the geometry of the galactic torsion axis relative to Earth's orbit. This interpretation remains strictly speculative.

8. Conclusions

This document presents a pre-registered, falsifiable protocol to search for a phase-drifting sidereal modulation in three existing neutrino datasets. The protocol specifies: fixed frequency f_{sig} = 2.3212e-05 Hz; blind amplitude scan; free residual drift parameter ε; unbinned MLE with externally constrained baselines; global significance via 10^5 timestamp-scrambled surrogates; three independent veto tests; random seed 20260414. The triple-lock fingerprint is frequency + residual drift + cross-dataset coherence. A positive detection requires all three acceptance criteria simultaneously. A null result publishes an upper bound $A < 10^{-4}$ (90% CL) in the neutrino sector — itself a new scientific result. This is a community re-analysis challenge. The data are already in the public domain. The data will decide.

OMEGA SERIES VOL. I — THE KINETIC HISS: SUMMARY

Frequency: f_sig = 2.3212e-05 Hz (period 11.9672h = 2*omega_sid)

Amplitude: A scanned blindly over [10^-5, 10^-3]

Residual drift: eps scanned over [-0.2, +0.2] s/day | eps_0 = 0.12 s/day (hypothesis)

MLE pipeline: unbinned, external-only baseline, Minuit2 MIGRAD

Significance: 10^5 surrogate scrambles | 5-sigma: p < 2.87e-7

Three vetoes: solar null + phase coherence + timestamp scramble

Triple-lock: frequency + drift + cross-dataset coherence

Baksan 24yr: phase-drift lever arm | SNO+: amplitude precision | IceCube: directional

Random seed: 20260414 | Protocol frozen at submission

The data will decide.

References

[1] Boliev M.M. et al. (Baksan). JCAP 09, 019 (2013). arXiv:1301.1138

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